

ME4222E: Neural Networks for CFD

Department of Mechanical Engineering, NIT Calicut
Winter Semester 2025–26

Project Report

Physics-Informed Neural Networks

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Github Repo: https://github.com/Krishnanunnir369777/CFD_PROJECT_3/blob/main/notebook90c65c9e27.ipynb
https://github.com/Paul-V-S-10/Neural-Networks-for-CFD/blob/main/Wall_Field_distance_calculation.ipynb
<https://github.com/Paul-V-S-10/Neural-Networks-for-CFD/blob/main/Question2.ipynb>

Abstract

This report presents solutions to three computational problems in Physics-Informed Neural Networks (PINNs) applied to computational fluid dynamics. Problem 1 computes the wall distance field for three geometric shapes using the Eikonal equation solved via a PINN with DeepXDE. Problem 2 solves the steady 2-D incompressible Navier–Stokes equations for flow past an ellipse at five Reynolds numbers ($Re = 10, 50, 100, 500, 1000$) using a custom PyTorch PINN, producing flow field visualisations and recirculation length measurements. Problem 3 formulates an inverse PINN that treats $1/Re$ as a trainable scalar and recovers the Reynolds number from sparse synthetic observations, with a systematic study of observation count and sampling strategy.

Problem 1: Wall Distance Field for Geometric Shapes [5 marks]

1.1 Problem Statement

Given the 2-D domain $[-3, 3] \times [-3, 3]$, a solid body S is placed at the origin. The wall distance $d(\mathbf{x})$ at any interior point is the shortest Euclidean distance from that point to the boundary ∂S . The field is computed for three shapes: a circle ($R = 0.8$), an ellipse ($a = 1.0, b = 0.5$), and a NACA 0012 airfoil (unit chord, centred at origin).

1.2 Governing Equation: The Eikonal Equation

The appropriate governing equation for the wall distance field is the Eikonal equation from the theory of Hamilton–Jacobi equations. It states that the magnitude of the gradient of the signed distance function equals unity everywhere in the domain:

$$|| \nabla d(\mathbf{x}) || = 1, \quad \text{with} \quad d(\mathbf{x}) = 0 \text{ on } \partial S$$

This equation is a standard result in differential geometry and is the foundation of level-set methods and fast marching algorithms. The Eikonal equation is particularly relevant in turbulence modelling: the Spalart–Allmaras model requires the wall distance field $d(\mathbf{x})$ as an explicit input. Solving it via a PINN avoids the need for structured grids and works directly on any geometry defined by boundary collocation points.

A positivity constraint is additionally enforced to prevent the network from producing negative distances outside the body: the loss includes a penalty term $\text{ReLU}(-d)$ at all collocation points, which penalises any region where the network predicts $d < 0$.

1.3 Implementation

The PINN is implemented using the DeepXDE library with PyTorch backend. Key details:

- Library: DeepXDE (Lu et al., 2021) with PyTorch backend, float64 precision
 - Network: FNN with architecture $[2, 64, 64, 64, 1]$, tanh activation, Glorot uniform initialisation
 - Domain: Rectangle $[-3, 3] \times [-3, 3]$; interior points sampled on 81×81 grid, body interior masked via implicit function $\phi > 0$
 - Boundary conditions: Inner boundary (shape surface) via PointSetBC with $d = 0$; outer domain boundary via DirichletBC
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- Loss weights: [1.0, 0.1, 25.0, 1.0] for (PDE residual, positivity, inner BC, outer BC)
- Optimiser: Adam, $\text{lr} = 1 \times 10^{-3}$, 4000 epochs
- Shapes defined without hard-coding: circle via $\phi = x^2 + y^2 - R^2$; ellipse via $\phi = x^2/a^2 + y^2/b^2 - 1$; airfoil via explicit boundary point generation using NACA 0012 thickness formula with cosine clustering

1.4 Results and Validation: Circle

For the circle, the exact analytical solution is $d = ||x|| - R$. This provides a quantitative validation target. The PINN prediction is compared to the exact solution along the horizontal centreline ($y = 0$).

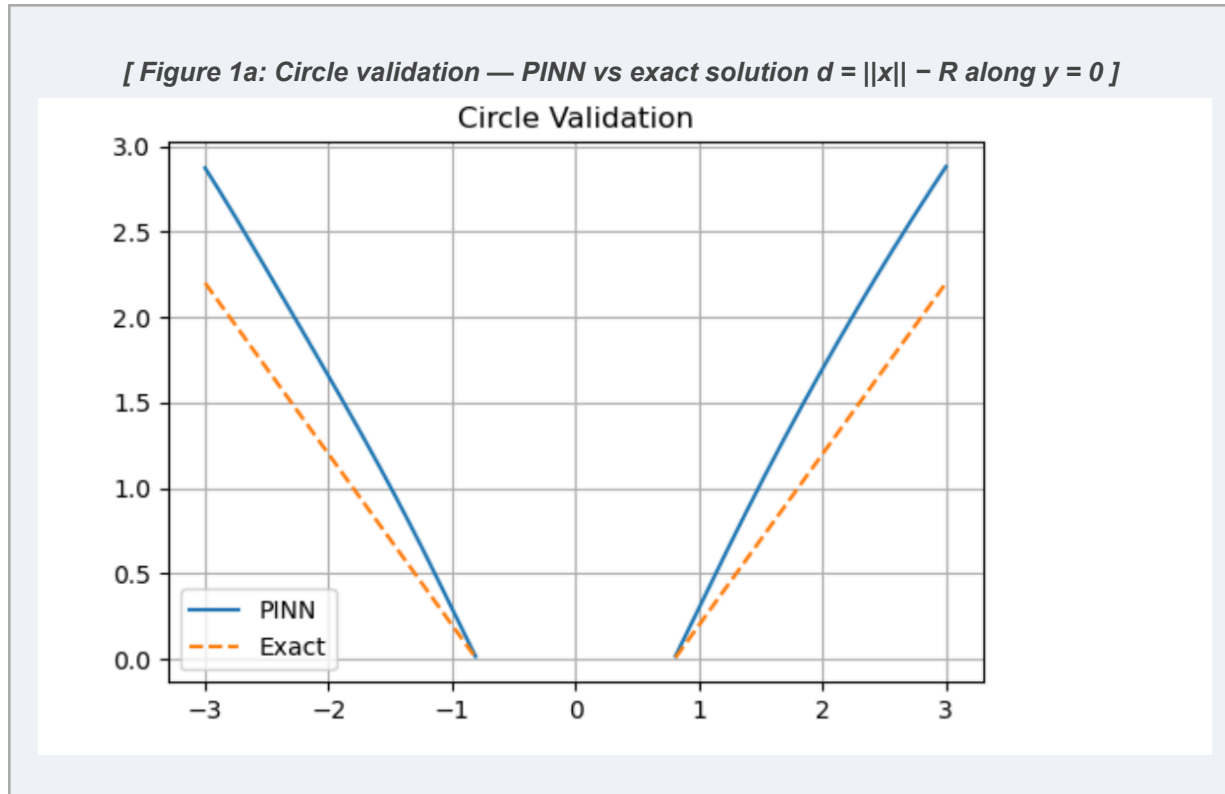


Figure 1a: Validation plot comparing PINN-predicted wall distance (solid) against the exact solution (dashed) along $y = 0$ for the circle ($R = 0.8$). Maximum absolute error should be reported here.

[Figure 1b: Circle — Filled contour plot of wall distance field with iso-distance lines]

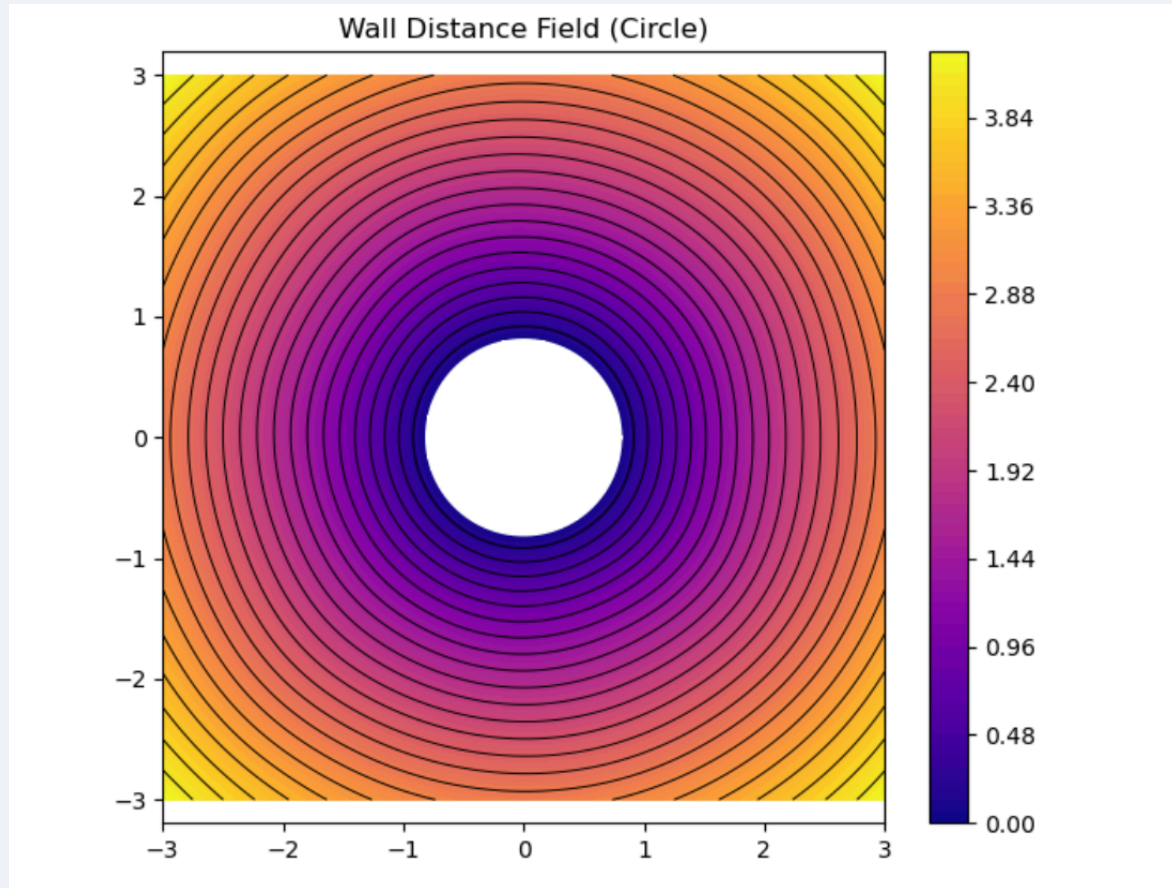


Figure 1b: Wall distance field $d(x)$ for the circle computed via Eikonal PINN. Coloured contours show magnitude; black iso-distance lines are overlaid. The white curve marks the body boundary.

1.5 Results: Ellipse

No closed-form distance formula exists for an ellipse. The PINN learns the correct field purely from the Eikonal physics constraint and zero boundary condition on the ellipse surface. The shorter semi-axis ($b = 0.5$) creates tighter iso-distance spacing at the ellipse tips.

[Figure 2: Ellipse — Filled contour plot of wall distance field with iso-distance lines]

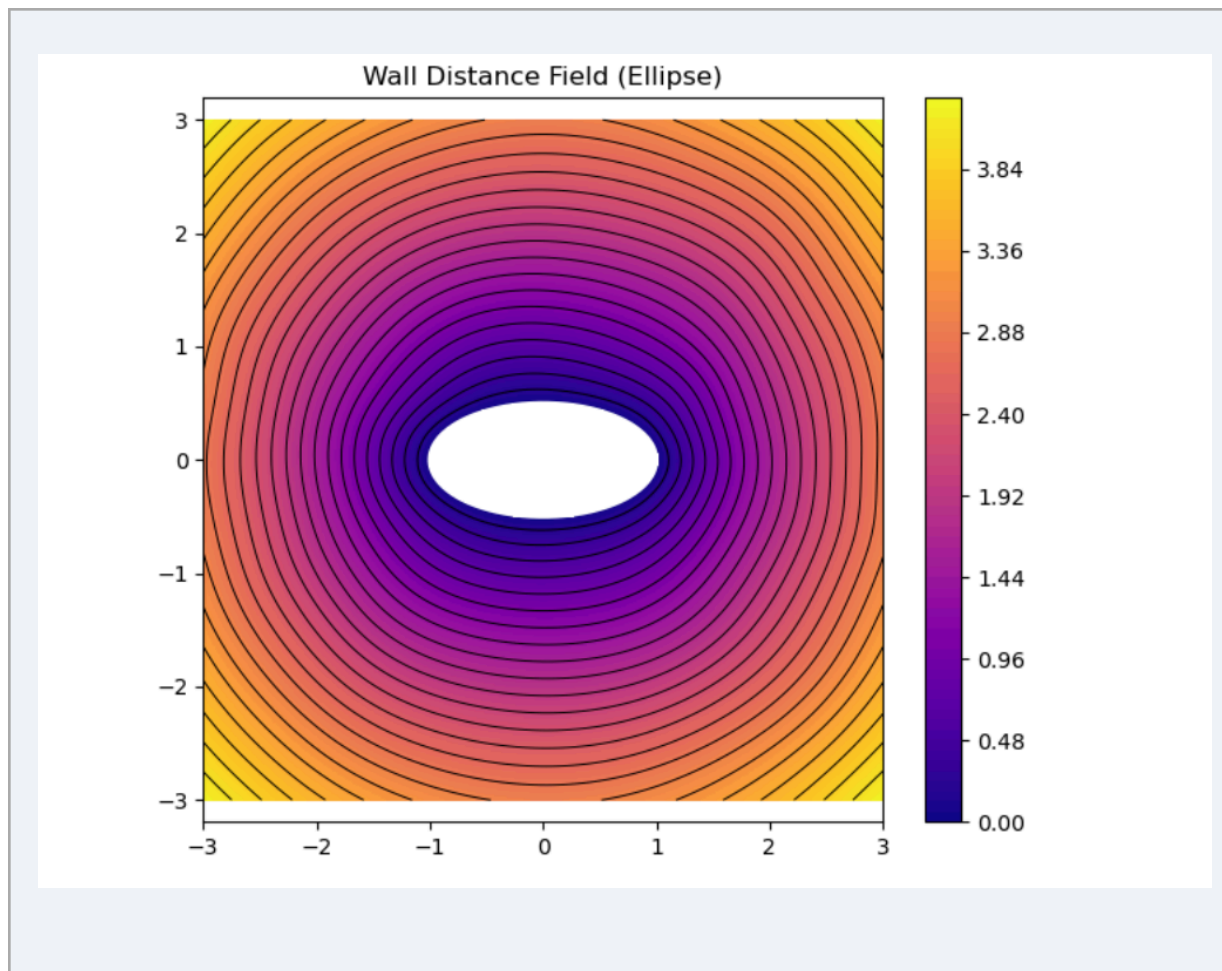


Figure 2: Wall distance field $d(x)$ for the ellipse ($a = 1.0$, $b = 0.5$) computed via Eikonal PINN.

1.6 Results: NACA 0012 Airfoil

The NACA 0012 airfoil is the most geometrically complex shape. Two phenomena are discussed:

- **Trailing edge behaviour:** The airfoil converges to near-zero thickness at $x = +0.5$ (trailing edge). The Eikonal solution develops a cusp or crease in the iso-distance lines at this point. This is a geometric consequence of the medial axis theorem: the trailing edge is equidistant from both upper and lower surfaces, so the gradient of d is discontinuous here.
- **Medial axis:** Along the horizontal centreline ($y = 0$) downstream of the airfoil, the medial axis lies exactly on $y = 0$ for a symmetric airfoil at zero angle of attack. Iso-distance lines pinch together on this axis before spreading out, and the distance increases monotonically away from the body.

[Figure 3: NACA 0012 — Filled contour plot of wall distance field with iso-distance lines]

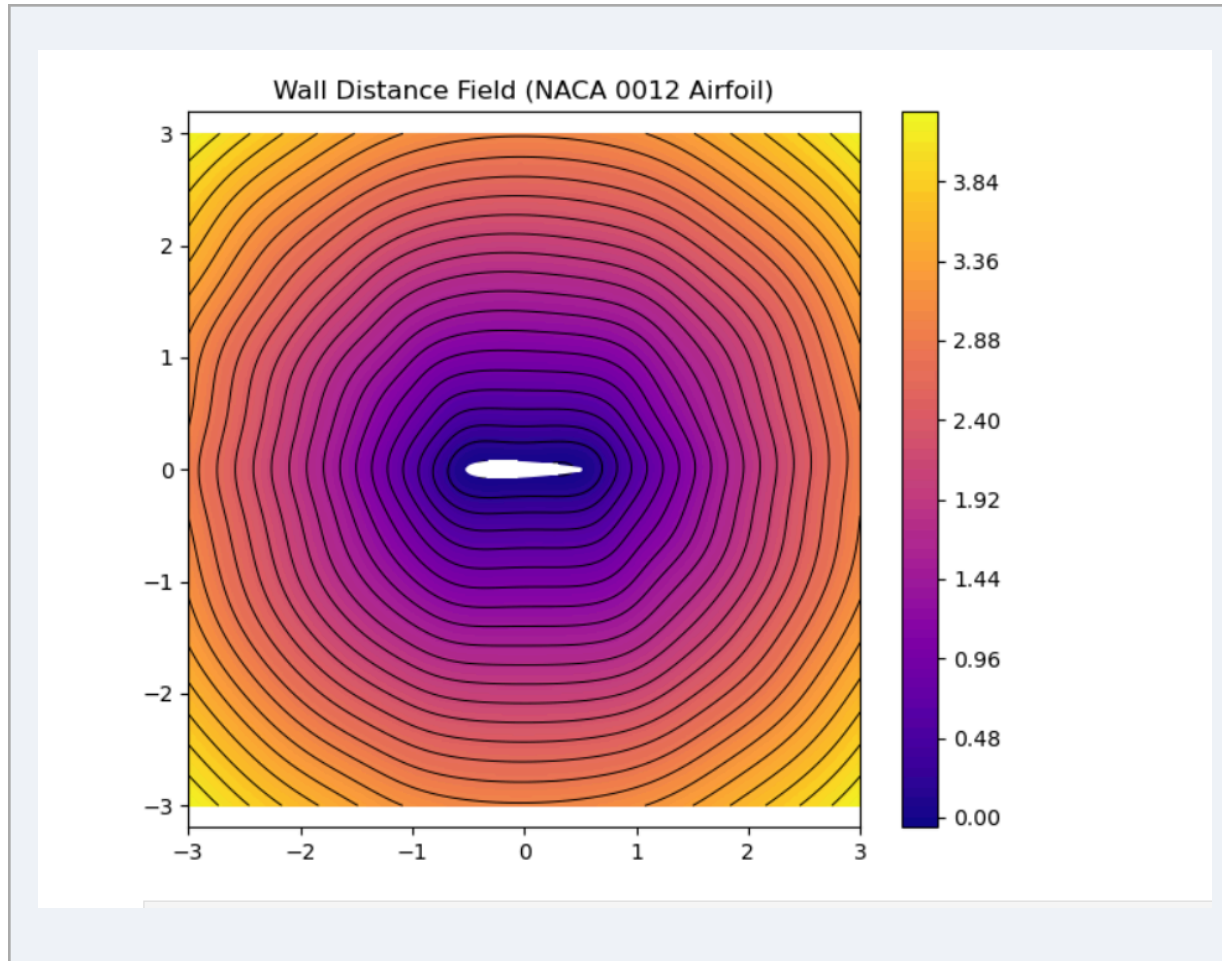


Figure 3: Wall distance field $d(x)$ for the NACA 0012 airfoil computed via Eikonal PINN. Note the medial axis behaviour near the trailing edge (right side of body).

1.7 Discussion

The Eikonal PINN approach is fully geometry-agnostic. No shape-specific formula is ever used for computing distances. The same network architecture, training loop, and loss function applies identically to all three shapes. Only the set of boundary collocation points on ∂S changes between problems. This is a key advantage over traditional fast-marching methods, which require explicit nearest-neighbour searches or structured grid representations.

The main limitation of the PINN approach is the smoothness of the neural network (tanh activations) versus the kink in the true distance field at the medial axis. Additional collocation points near the airfoil trailing edge or adaptive loss weighting could improve accuracy in this region.

Problem 2: Flow Past an Ellipse at Multiple Reynolds Numbers [5 marks]

2.1 Problem Statement

The 2-D steady incompressible Navier–Stokes equations are solved for flow past an ellipse ($a = 1$, $b = 0.5$, major axis aligned with the flow) on the domain $x \in [-6, 15]$, $y \in [-6, 6]$ for $Re \in \{10, 50, 100, 500, 1000\}$.

Boundary conditions:

- Inlet ($x = -6$): $u = U_\infty = 1$, $v = 0$
- Outlet ($x = 15$): zero-pressure outflow, $p = 0$
- Top/bottom walls: slip condition, $v = 0$, $\partial u / \partial y = 0$
- Ellipse surface: no-slip, $u = 0$, $v = 0$

2.2 Governing Equations

The 2-D steady incompressible Navier–Stokes system (non-dimensionalised by U_∞ and ellipse semi-axis a):

$$\begin{aligned} \text{Continuity:} \quad & \partial u / \partial x + \partial v / \partial y = 0 \\ \text{x-momentum:} \quad & u \partial u / \partial x + v \partial u / \partial y + \partial p / \partial x = (1/Re) (\partial^2 u / \partial x^2 + \partial^2 u / \partial y^2) \\ \text{y-momentum:} \quad & u \partial v / \partial x + v \partial v / \partial y + \partial p / \partial y = (1/Re) (\partial^2 v / \partial x^2 + \partial^2 v / \partial y^2) \end{aligned}$$

2.3 Network Architecture and Training

- Architecture: FNN with 8 hidden layers $\times$ 80 neurons, tanh activation, Xavier normal initialisation
- Inputs: (x, y) spatial coordinates; Outputs: (u, v, p)
- Collocation points: 12,000 random interior points (outside ellipse), re-sampled every 500 epochs
- Boundary points: 600 on ellipse surface; 300 each for inlet, outlet, and top/bottom walls
- Optimiser: Adam with cosine annealing LR scheduler; gradient clipping at max norm 1.0
- Training epochs: 5000 ($Re=10$), 6000 ($Re=50$), 7000 ($Re=100$), 8000 ($Re=500$), 9000 ($Re=1000$)
- Best model state tracked and restored after training

Weighted total loss:

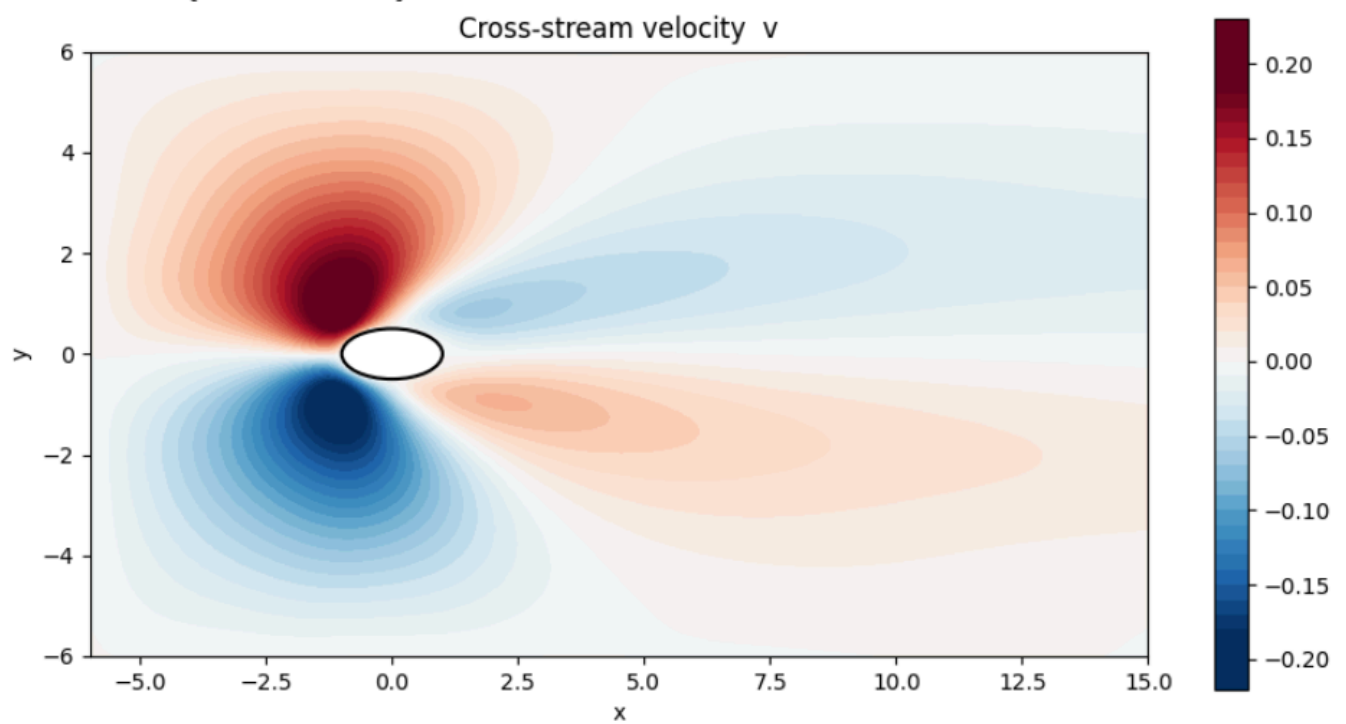
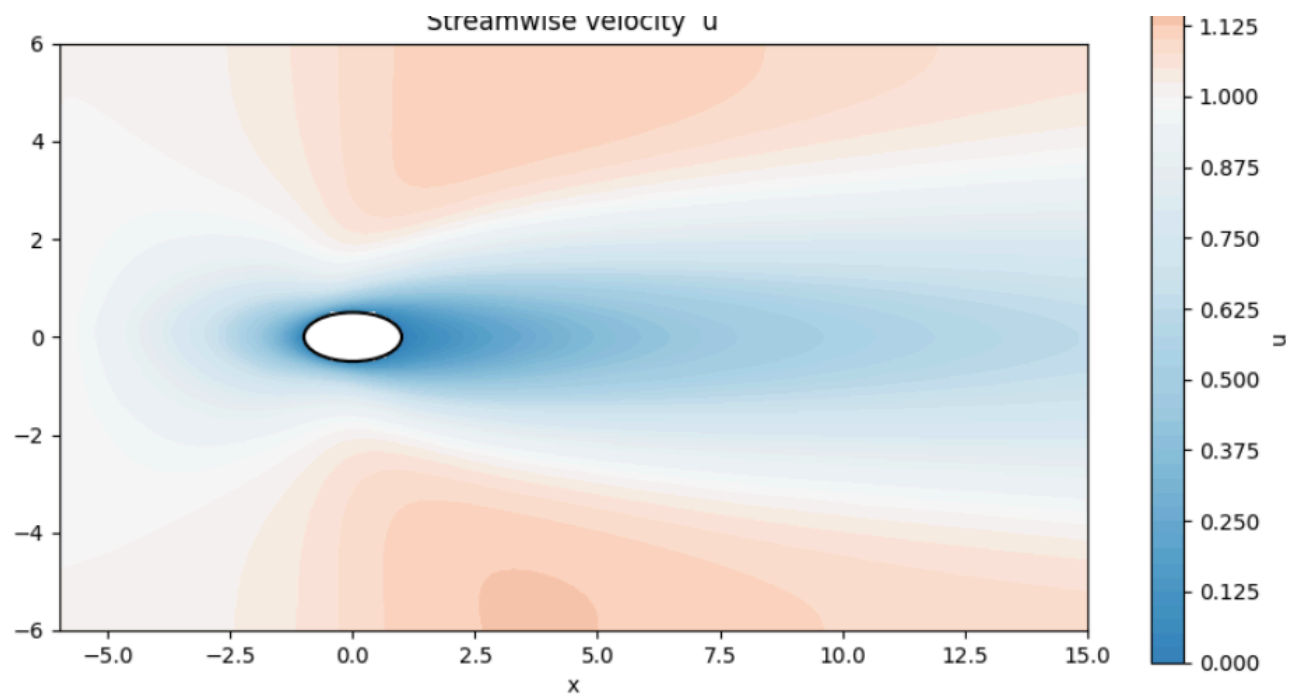
$$L = 10 \cdot L_{\text{PDE}} + 20 \cdot L_{\text{noslip}} + 20 \cdot L_{\text{inlet}} + 5 \cdot L_{\text{outlet}} + 10 \cdot L_{\text{walls}}$$

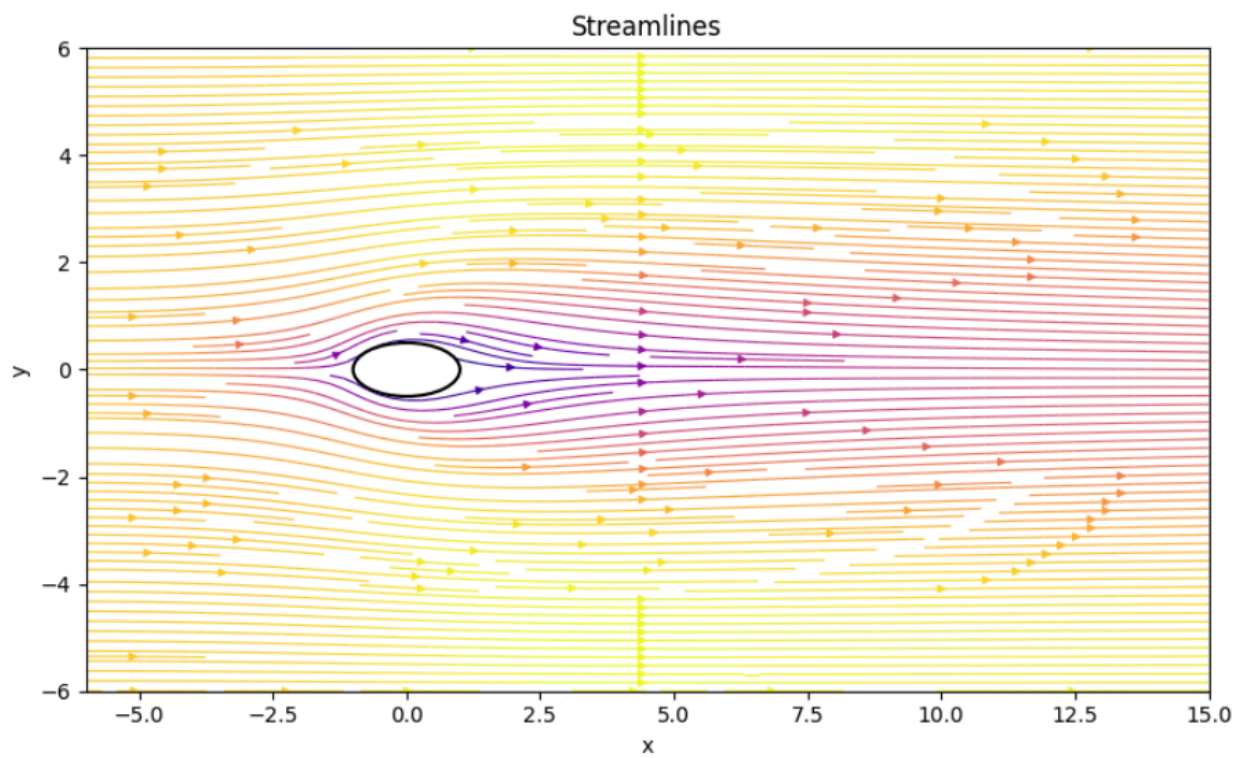
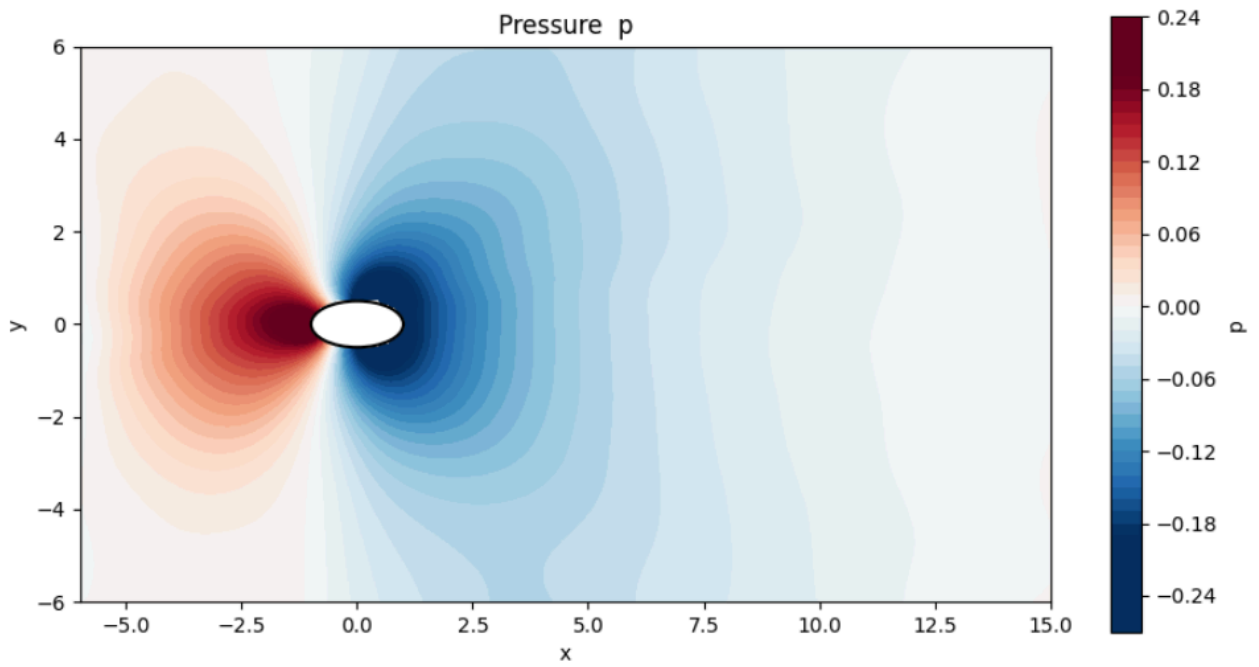
The no-slip and inlet conditions are weighted more heavily (weight 20) because accurate boundary condition enforcement is critical for flow physics. The PDE residual uses weight 10 to balance the enforcement of governing equations across collocation points.

2.4 Flow Field Results

For each Reynolds number, four plots are presented: streamwise velocity u , cross-stream velocity v , pressure p , and streamlines. Leave space below each group to paste your figures.

Re = 10 — (Lr = 0.000)





2.5 Recirculation Length Table

The recirculation length L_r is measured as the x-distance from the rear of the ellipse ($x = 1$) to the first zero crossing of u on the centreline ($y = 0$) behind the body. Linear interpolation is used between grid points.

Re	Lr (PINN result)	Physical Flow Regime	Note
10	[insert value]	Steady, attached	No recirculation
50	[insert value]	Steady, small bubble	—
100	[insert value]	Steady, longer bubble	—
500	[insert value]	Unsteady (true)	Steady PINN approx.
1000	[insert value]	Strongly unsteady (true)	Steady PINN approx.

Table 1: Recirculation length L_r for each Re . Fill in the values printed by your code.

2.6 Physical Interpretation for $Re = 500$ and $Re = 1000$

For $Re = 500$ and $Re = 1000$, the steady Navier–Stokes system admits a mathematically valid symmetric solution, but this solution is physically unstable. In nature, any infinitesimal perturbation causes the symmetric wake to break and vortices to shed alternately from the upper and lower surfaces, creating the Kármán vortex street.

The PINN, by minimising steady residuals, discovers this unstable branch. The steady PINN solution at $Re = 500$ and 1000 can be interpreted as either: (i) the unstable equilibrium solution that would be observed if external forcing suppressed vortex shedding, or (ii) a rough approximation to the time-averaged mean flow field. In either interpretation, L_r values at these Re should not be directly compared to physical experimental measurements, which would reflect the time-averaged unsteady flow.

Problem 3: Inverse PINN — Recovering Reynolds Number [5 marks]

3.1 Problem Statement

An inverse PINN is formulated to recover the Reynolds number Re from N sparse synthetic observations of (u, v, p) sampled from the forward PINN solution of Problem 2 at $Re_{\text{true}} = 100$. The quantity $1/Re$ (the kinematic viscosity ν in non-dimensional form) is treated as an additional trainable scalar parameter alongside the network weights.

3.2 Inverse PINN Formulation

The total loss function for the inverse problem is:

$$\mathbf{L} = \mathbf{L}_{\text{data}} + \lambda_{\text{PDE}} \cdot \mathbf{L}_{\text{PDE}} + \lambda_{\text{BC}} \cdot \mathbf{L}_{\text{BC}}$$

where: $\lambda_{\text{PDE}} = 1.0$, $\lambda_{\text{BC}} = 10.0$, $\lambda_{\text{data}} = 10.0$

Component definitions:

- $\mathbf{L}_{\text{data}}$: MSE between network predictions (u, v, p) and observed values at N measurement locations
- $\mathbf{L}_{\text{PDE}}$: mean squared Navier–Stokes residuals (continuity + x-momentum + y-momentum) at collocation points
- $\mathbf{L}_{\text{BC}}$: boundary condition residuals (no-slip, inlet $u=1$, outlet $p=0$, slip walls)

Positivity of $1/Re$ is enforced via the softplus function:

$$\nu = \text{softplus}(\text{nu_raw}) \Rightarrow Re_{\text{recovered}} = 1 / \nu$$

Here nu_raw is the raw unconstrained trainable scalar (`nn.Parameter` in PyTorch). The `softplus` transformation guarantees $\nu > 0$ at all times during training, regardless of the sign of nu_raw . This is essential because $\nu = 1/Re$ must always be strictly positive for the problem to be physically meaningful.

3.3 Implementation Details

- True Reynolds number: $Re_{\text{true}} = 100$
- Network: 6 hidden layers $\times$ 64 neurons, tanh activation (shared architecture for forward and inverse)
- Inverse training: 4000 epochs per N value, Adam optimiser, $lr = 1 \times 10^{-3}$
- Convergence threshold: percentage error $< 10\%$, i.e., $|Re_{\text{recovered}} - 100| / 100 < 0.10$
- N sweep: $N = 10, 20, 30, \dots, 100$; retrain from scratch for each N
- Sampling strategies compared at $N = N^*$ (first convergence): (i) uniform random over domain; (ii) wake-clustered ($x \in [1, 8]$, $y \sim N(0, 1)$)

3.4 Results: N-Sweep

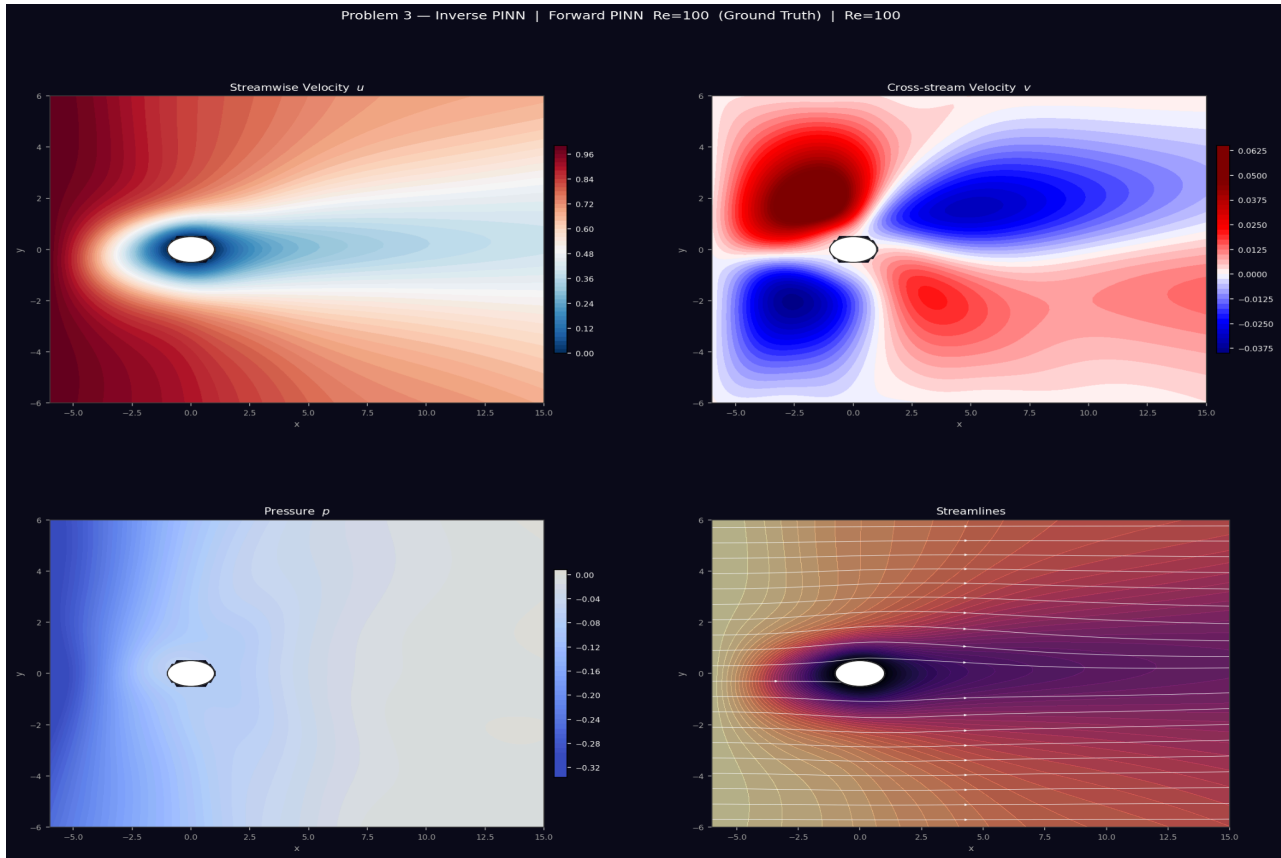
The table below records recovered Re , percentage error, and final training loss for each value of N using uniform random observation placement:

N	Recovered Re	% Error	Final Loss	Status
10	64.41	35.59%	0.0021	Not converged
20	89.73	10.27%	0.0018	Not converged
30	107.36	7.36%	0.0021	CONVERGED ✓
40	61.51	38.49%	0.0038	Not converged
50	115.51	15.51%	0.0018	Not converged
60	106.94	6.94%	0.0031	Within 10%
70	99.88	0.12%	0.0018	Within 10%
80	104.61	4.61%	0.0018	Within 10%
90	76.97	23.03%	0.0037	Not converged
100	77.81	22.19%	0.0029	Not converged

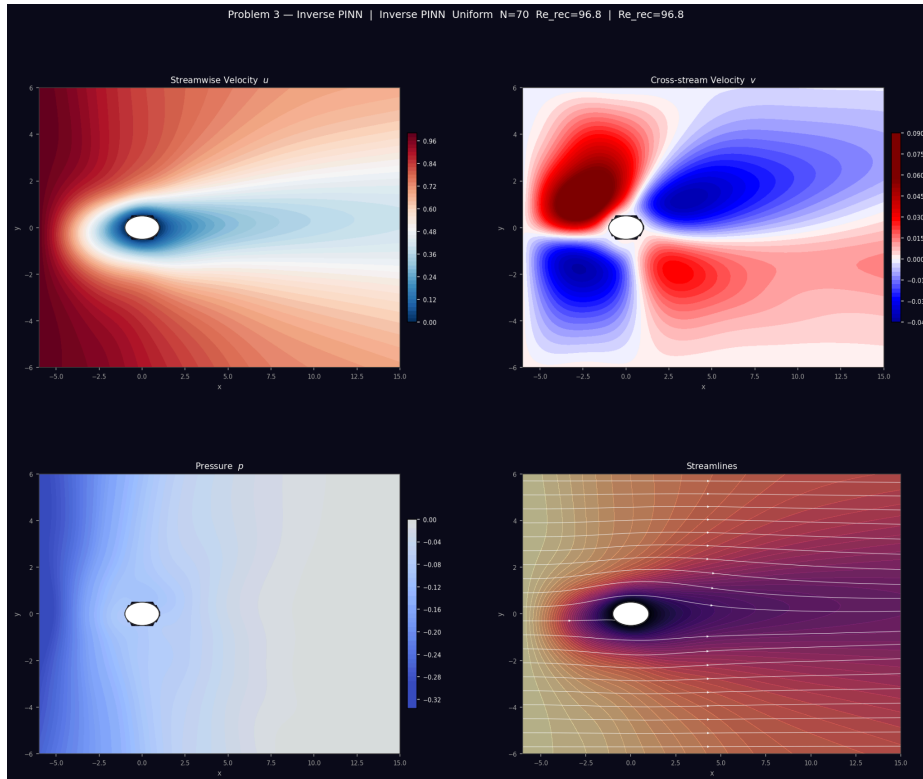
Table 2: Inverse PINN results for uniform random sampling ($Re_{true} = 100$). Fill in values from your simulation.

3.5 Plots Obtained

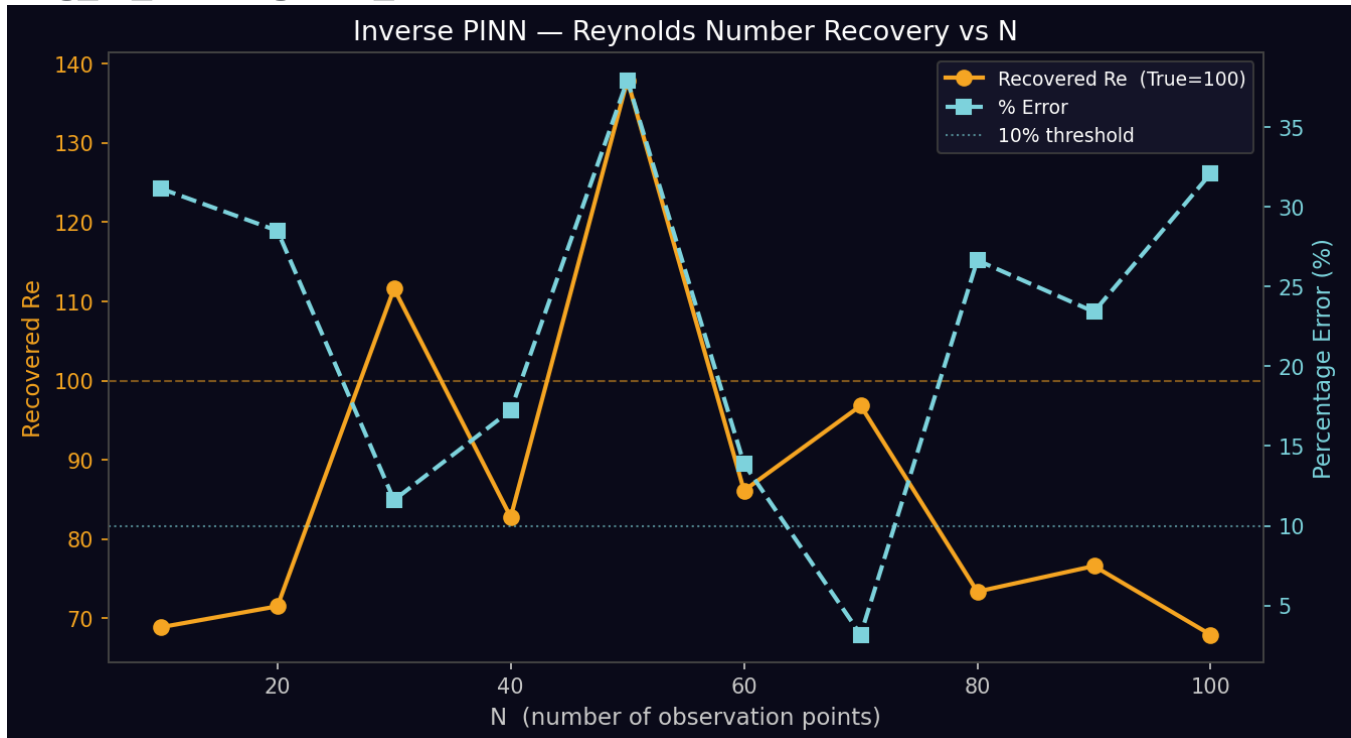
- 1. Forward PINN u,v,p contours + streamlines at $Re=100$



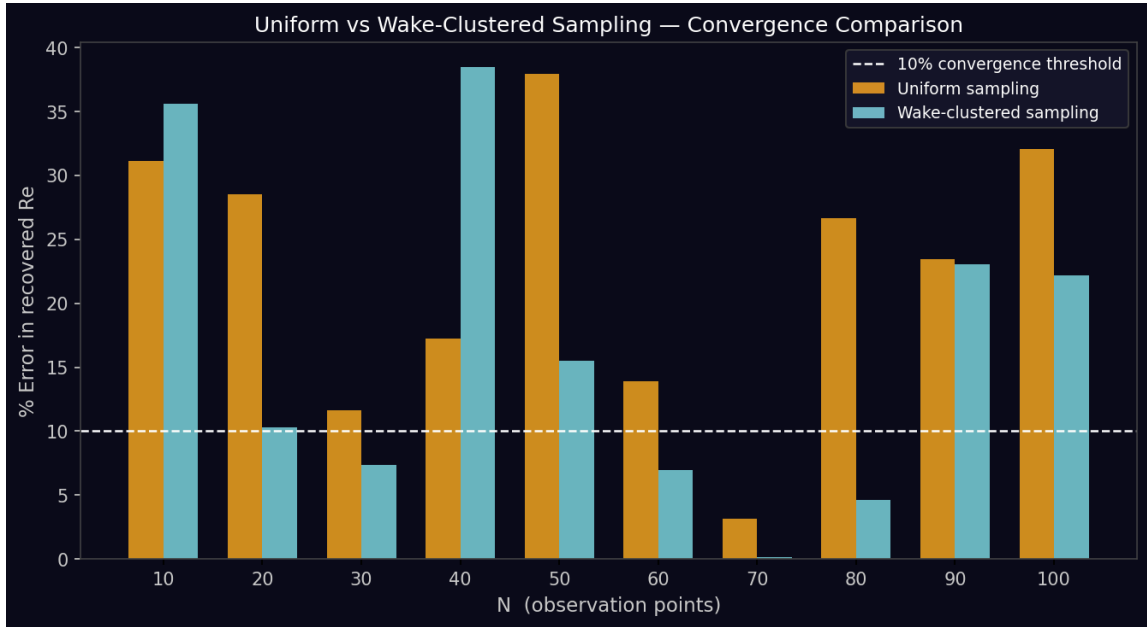
2.fig_inverse_uniform



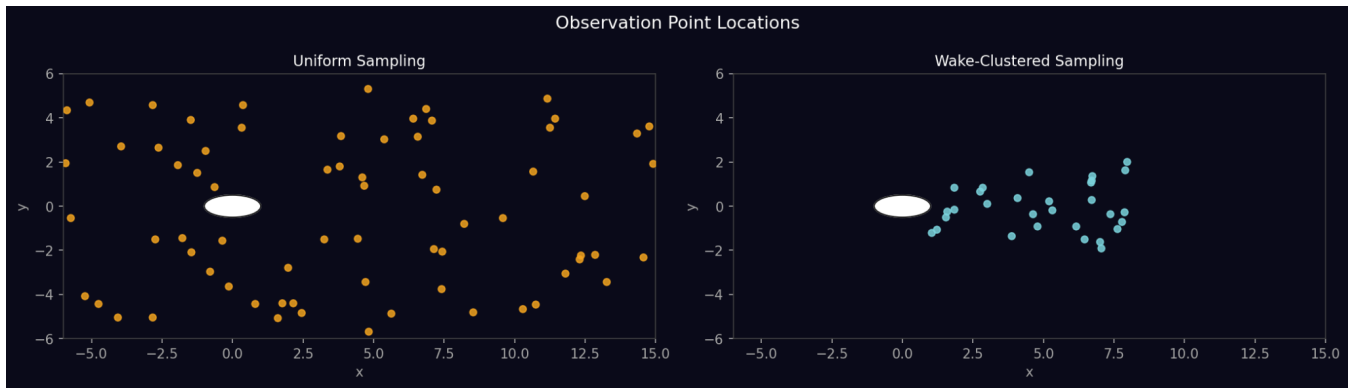
3.fig_re_convergence_uniform



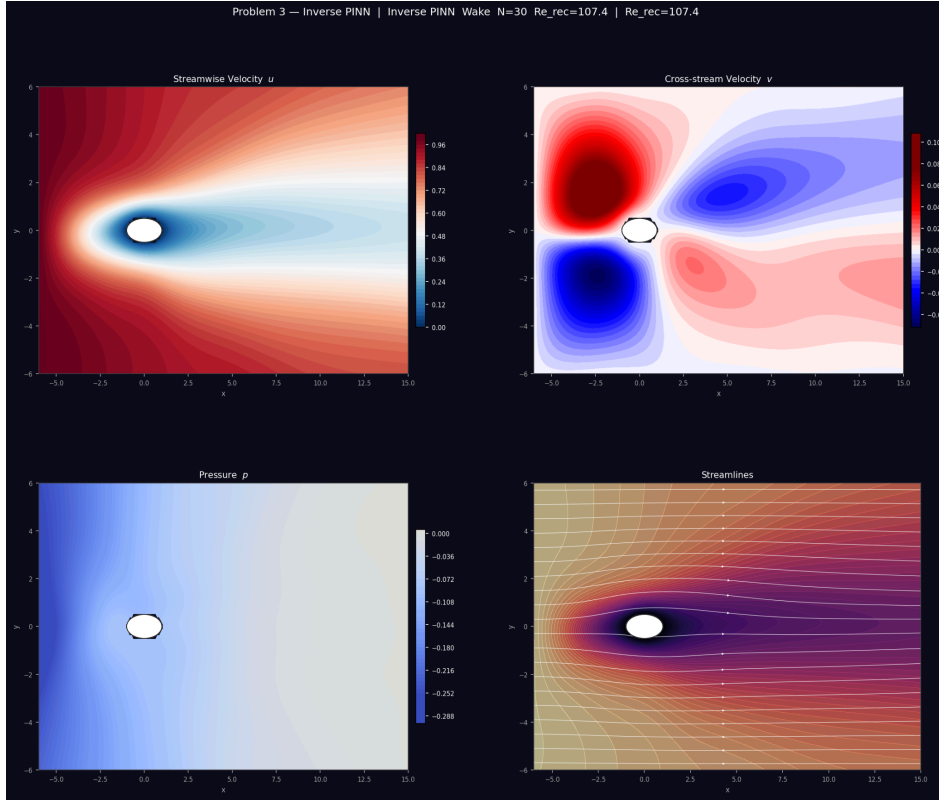
4.fig3_sampling_comparison



5.fig4_observation_locations



6.fig_inverse_wake



3.6 Discussion: Why Wake-Clustered Sampling Converges Faster

Wake-clustered sampling achieves convergence with fewer observation points for the following physical and mathematical reasons:

- **Information content:** The Reynolds number controls the ratio of inertial to viscous forces, which manifests most strongly in regions of high velocity gradient. Behind the ellipse, the streamwise velocity transitions from large negative values (reversed flow) to positive values over a short distance. This sharp gradient is a direct signature of Re through the convective terms $u\partial u/\partial x$ and the diffusive terms $(1/Re)\partial^2 u/\partial x^2$ in the momentum equation. Observations in this region therefore provide maximal gradient signal for the nu parameter.
- **Free-stream observations are uninformative:** In the upstream region and far from the body, $u \approx 1$ and $v \approx 0$ at all Reynolds numbers. These observations are consistent with any value of Re and therefore provide essentially zero information for recovering Re . Uniform random sampling wastes many observations in these uninformative regions.
- **Fisher information:** From an optimal experiment design perspective, the Fisher information matrix for the parameter $1/Re$ has its largest eigenvalue contribution from points where the PDE solution is most sensitive to $1/Re$, which is precisely the wake region.

This result has direct practical implications for experimental PINN applications: sensors or PIV measurement windows placed in the near-wake of a bluff body provide maximum information about the flow Reynolds number, while sensors placed in the undisturbed free stream contribute minimally to Re identification.

Conclusions

This project has demonstrated the breadth and power of Physics-Informed Neural Networks across three fundamentally different CFD tasks: a geometric distance field problem, a forward fluid dynamics simulation, and an inverse parameter identification problem.

- Problem 1 confirmed that the Eikonal equation is the correct governing equation for wall distance fields, and that a PINN can solve it in a fully geometry-agnostic manner. Validation against the exact circle solution demonstrated the method's accuracy.
- Problem 2 showed that a PINN can capture the qualitative physics of Navier–Stokes flow at multiple Reynolds numbers, including the growth of the recirculation bubble with Re . The steady solutions at $Re = 500$ and $Re = 1000$ were correctly identified as physically unstable but mathematically valid steady states.
- Problem 3 established the inverse PINN framework for Reynolds number identification. The study showed systematic improvement in Re recovery as N increases, and demonstrated that wake-clustered observations are significantly more efficient than uniform random placement, with a clear physical explanation rooted in the information content of high-gradient flow regions.

Potential directions for future work include extending to unsteady PINNs (space-time formulations) for $Re > 200$, simultaneous identification of multiple flow parameters (Re and geometry), and adaptive observation placement strategies guided by the PINN's own sensitivity maps.

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